

Split ↓	# Questions	# Tables	# Pages	Popularity	Sensitivity
Train	10,148	6,109	5,751	10912.7	14.6
Dev.	1,000	582	539	5592.2	16.6
Test	9,000	5,290	4,725	6812.5	16.9

Table 1: Statistics of TAQA dataset. We use Wikipedia pages’ averaged monthly pageview to measure questions’ popularity, and sensitivity is defined as Equation 2. Popularity and sensitivity are averages across examples in each split.

be kept in our TAQA dataset.

Train-test split. Finally, we randomly split the 20,148 questions into train, development, and test sets in TAQA. To better probe LMs’ understanding of temporal knowledge like Figure 1, we only put questions that can be answered since 2000 into the dev. and test sets. We also ensure that there is no QA-originated Wikipedia page overlap between the train, dev., and test sets, as we want to avoid LMs improving their QA performance by memorizing the training data’s knowledge. The data characteristics are illustrated in Table 1.

Quality control. To examine the overall quality of the generated TAQA dataset, we randomly sampled 200 questions along with their answers and manually verified their correctness. Among the 200 examples, 185 (92.5%) instances are considered correct, while the incorrectly generated questions are mostly because tables without temporal information are used for QA generation (6/200, 3.0%). Another major issue is that GPT-4 sometimes automatically adds conditional clauses to the question, thus making them not time-sensitive anymore (5/200, 2.5%).

Dataset characteristics In Table 2, we show a detailed comparison between TAQA and existing time-sensitive QA datasets. Overall, TAQA maintains more time-sensitive questions than existing datasets, making it sufficient for training LMs. Furthermore, questions in TAQA are more sensitive to time than existing datasets, and most answers to questions (95.9%) were changed after 2019 when modern LMs were trained. TAQA also tracks time-sensitive questions over a long period of time (21.5 years on average). Figure 3 also shows that TAQA contains more questions that have corresponding answers recorded after 2000³. These properties

³The Wikipedia tables we use and our intermediate data of TAQA contain answers before 2000 and after 2023, but we

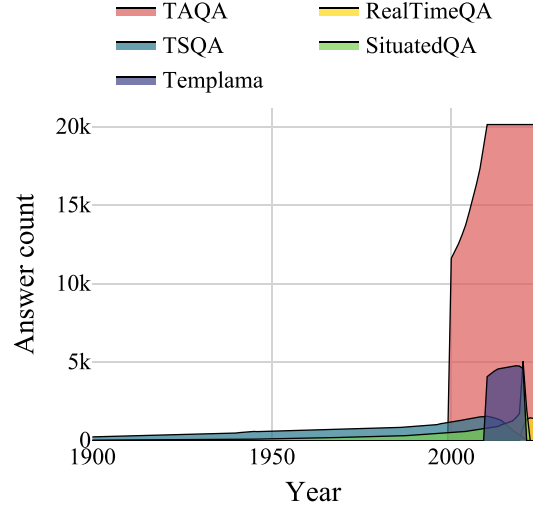


Figure 3: The comparison of TAQA and existing time-sensitive QA datasets regarding the frequency of the answers’ corresponding time. TAQA contains more data in the post-2000 era.

make TAQA suitable for diagnosing modern LMs’ understanding of time.

2.3 Evaluation Metrics

In TAQA, we focus on the question-answering task and expect models to output short answers. So, following prior QA evaluation (Rajpurkar et al., 2016; Kwiatkowski et al., 2019), we adopt the token-level F1 score to measure the quality of the LM-generated answers given ground truth answers. We further developed three variants tailored for the temporal alignment evaluation.

Target-year F1 Since the ground truth of a question q in our TAQA dataset changes across time, we first calculate the F1 score of an LM-predicted answer $\hat{a}$ to the ground truth in each year between 2000 to 2023, for example, $\mathcal{F}^{2023}(\hat{a}) = \text{F1}(\hat{a}, a_{2023})$ for the LM-predicted answer’s accuracy as of in 2023.

Max F1 over history To determine whether an LM can answer questions with historically correct answers if the recent knowledge is unknown to the LM, we also calculate the maximum F1 score among different years’ ground truth answers, i.e., $\mathcal{F}_{\max}(\hat{a}) = \max_{2000 \leq i \leq 2023} \mathcal{F}^i(\hat{a})$.

Decayed F1 towards target Ideally, temporally aligned LMs’ $\mathcal{F}_{\max}$ scores should not be worse than

discard them for simplicity as these answers are not used in LM temporal alignment analysis.

Dataset	# Questions	# Post-2019 answer change	Time	Median year	Track length	Avg. sensitivity
SituatedQA (Zhang and Choi, 2021)	5,061	1,523	1270–2021	2005	8.5	1.7
Templama (Dhingra et al., 2022b)	5,798	1,082	2010–2020	2015	8.9	3.3
TSQA (Chen et al., 2021)	5,061	134	1–2021	1963	26.0	4.4
RealtimeQA (Kasai et al., 2022)	5,152	0	2020–2024	2022	1.0	1.0
TAQA (ours)	20,148	19,316	2000–2023	2013	21.5	15.6

Table 2: Detailed dataset characteristics comparison between TAQA and existing time-sensitive QA datasets. Track length is defined as the average number of years a question’s answer is recorded.

the unaligned one. Furthermore, to differentiate the answers based on recent and outdated knowledge, we softly penalize the outdated answers by calculating the maximum value among F1 scores exponentially decayed by the time gap of the ground truths’ year i to the target year j , defined as:

$$\mathcal{F}_{\text{decay}}^j(\hat{a}) = \max_{2000 \leq i \leq 2023} (\mathcal{F}^i(\hat{a}) \cdot \alpha^{|i-j|}) \quad (3)$$

where $0 < \alpha < 1$ is the decaying factor, which controls acceptance of the outdated answers. When $\alpha = 1$, $\mathcal{F}_{\text{decay}}^j$ is the same as $\mathcal{F}_{\text{max}}$, and when $\alpha \rightarrow 0$, $\mathcal{F}_{\text{decay}}^j \rightarrow \mathcal{F}^j$. In practice, we set $\alpha = 0.8$ so that a three-year-outdated answer will receive only half of its original F1 score.

3 Aligning LMs towards Recency

In this section, we propose several methods for aligning LMs to recent years. Complementary to existing studies that aim to teach language models up-to-date knowledge (Jin et al., 2022; Meng et al., 2023), our alignment methods focus on how to steer models to reorganize their knowledge and align them to the most-recent possible time.

3.1 Time-Aware Prompting

An intuitive method to make LMs respond based on a specific time is to prompt them explicitly with the time information. We can achieve this by including the target year information. In addition, we also find models can benefit from being prompted with a few demonstration examples that are time-sensitive and with answers in the target time, in a sense that it is inferring the year information from these demonstration answers. We analyze the effects of these elements in Appendix §A. Therefore, we propose this time-aware prompting with the year information and a few time-sensitive examples in the prompt to activate models’ knowledge of a specific year. Specifically, we append the prompt “as of year y , the answer is” at the end of each question q , where y is the target year we want the LM

to be aligned to. For the demonstration examples, we randomly sample 5 instances multiple times from the top-200 popular⁴ QA instances in TAQA training set as few-shot examples, and choose the five examples that achieve the best performance on TAQA development set.

The chief advantage of this method is its simplicity, requiring no model updates; it can be used with models that only offer API access. However, prompting does not change models’ internal state of time, and for many users, the additional information and few-shot examples are unnatural and add extra costs to inference.

3.2 Target-Year Finetuning

We next explore ways to align LMs temporally by finetuning their parameters. As we want to change the models’ internal sense of time, we choose not to add any temporal information in the context so that LMs can only generalize when they adjust parameters to model the time information. We refer readers to Appendix §B for the training details.

One critical factor for finetuning is to select the data that can align LMs effectively. In fact, much up-to-date knowledge may not be presented to LMs during pretraining or not be memorized successfully. Tuning LMs with the knowledge they do not know can increase the risk of hallucination, i.e., LMs are encouraged to respond to questions with answers not seen during pretraining⁵. To address this, we propose selecting training data based on the correctness of sampled answers from the LMs. Given a question, we use the time-aware prompting in §3.1 to first sample ten outputs from the to-be-finetuned LM and calculate whether any of

⁴Here, we use the QA’s originated Wikipedia page’s averaged monthly pageview from 2016 to 2023 to represent popularity.

⁵This hallucination encouraging effect was mentioned in <https://news.berkeley.edu/2023/04/24/berkeley-talks-chatgpt-developer-john-schulman> and then widely discussed by the community.

the ten samples can hit the ground truth answer. We select the top 5000 examples from the TAQA training set with the largest F1 overlaps between model samples and ground truth answers. By finetuning on question-answer pairs where the model already has some chance of giving the correct answer, we hope to steer it toward the target year rather than “teaching it” entirely new facts. We also explored other data selection methods based on popularity or model confidence, which are reported in §4.3

3.3 Temporal-Adaptive Finetuning

Finetuning to a year assumes the model should answer all questions based on that target time; it, however, fails to take into account a desired graceful degradation: when the LM doesn’t have access to the target-year answer, it should revert back to the most recent prior answer.

To achieve this goal, we propose a temporally adaptive finetuning technique, where we dynamically determine the most recent and proper target year for each question. In practice, given a question, we iterate from the pretraining cutoff year (2022 for LLaMa2) to earlier years; for each year, we try using the correctness-based approach described in §3.2 to sample ten outputs, compare with the ground truth answers; if the F1 overlap is larger than a certain threshold (0.7), we will use this year as the target year for this question. As a result, we can adaptively assign each question with its expected target year, and then train the model to output “Based on my latest knowledge for this question from year . . . , the answer is:” before answering the question. Hence, we hope that the LM can learn to pick the target year for each question based on internal knowledge.

4 Experiments

This section describes our experiments on aligning LLaMa2 using our proposed methods on the TAQA dataset. Since the pretraining cutoff of LLaMa2 is September 2022, we try aligning LLaMa2 to both 2022, which it has seen partially, and 2021, which it should have seen entirely. We first introduce several baselines (§4.1), present the effectiveness of aligning LLaMa2 (§4.2), ablate different data selection strategies (§4.3), and finally conduct various analyses to verify the improvement (§4.4).

4.1 Baselines

Unaligned models To demonstrate the performance of LMs after pretraining, we choose several

representative LMs, including LLaMa1-65B (Touvron et al., 2023a), LLaMa2-70B (Touvron et al., 2023b), and GPT-3 (Brown et al., 2020) (the “Davinci-002” engine using OpenAI’s API). To make them answer questions in the right format, we prompt them with five time-insensitive QA examples, presented in Appendix §A.

Normally aligned models To show the necessity of temporal alignment against other types of alignment in prior work, we additionally test ChatGPT (“gpt-3.5-turbo-0125” engine), which went through an RLHF process (Ouyang et al., 2022), and Tülu2-70B (Iverson et al., 2023), a LLaMa2 model that underwent supervised instruction tuning.

4.2 Results of Recency Alignment

We present the results of aligning LLaMa2 to the years 2021 and 2022 in Table 3, as well as the performance of baseline models. All the **temporally aligned models perform better than unaligned and normally aligned models** in both target years, suggesting the possibility and effectiveness of our proposed temporal alignment. Among all the methods, **finetuning to the target year performs better than just prompting**. Given that we do not include any time information in the prompt for finetuned models, this suggests that the model is learning to adjust its internal state to activate more of its memorized recent pretraining knowledge. The **benefit of adaptive finetuning is more on historical years**, which is reflected by its best performance on $\mathcal{F}_{\max}$ metric and further confirmed by its performance on a broader range of years shown in Figure 1.

Moreover, comparing the two target years, our models always perform better on 2021. This could be because LLaMa2 did not see all the information in 2022 but we also speculate the information in recent years is less exposed in the pretraining data, as is also shown in Figure 1. We leave the investigation into pretraining data as future work.

4.3 What Data is More Proper for Alignment?

We further analyze the effectiveness of our correctness-based data selection strategy (**Model Correctness**) described in §3.2. We compare it with full-data training, as well as four other selection strategies, including random selection (**Random**), selecting the most popular questions according to page views (**Popularity**), and selecting questions that the model is mostly confident about its answer (**Model Confidence**). All the se-